Design, Simulation and Physical Verification of a MOSFET-Based Widlar Current Mirror

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Academic Year 2026–2027, Semester VII

***Abstract*—This paper presents the design, simulation, layout and physical verification of an NMOS-based Widlar current mirror implemented in Cadence Virtuoso using the GPDK090 technology library. The circuit uses two equal NMOS transistors (W = 3.12 µm, L = 280 nm), a 100 µA reference current, a source-degeneration resistor realized as an on-chip non-salicided P+ polysilicon resistor, and a 1 MΩ output load at a 1.8 V supply. The target output current is 1 µA. A square-law approximation gives an initial source resistance of approximately 204 kΩ; a DC parametric sweep against the BSIM3v3 technology models selects approximately 271.0 kΩ, realized as a 28-segment, 677.46 µm serpentine poly resistor of 270.98 kΩ. The simulated output current is 999.97 nA (0.003% error). A full custom layout of the mirror was drawn, passing Diva DRC with zero errors and Diva LVS with a clean match against the schematic. The gap between the hand calculation and simulation, and the additional constraints imposed by physical design, highlight the importance of technology-model-based verification.**

***Index Terms*—Widlar current mirror, MOSFET, current reference, analog integrated circuits, Cadence Virtuoso, GPDK090, DRC, LVS.**

I. INTRODUCTION

Current mirrors are widely used in analog integrated circuits to establish bias currents for amplifiers and other circuit blocks. In a basic MOS current mirror, a diode-connected reference transistor establishes a gate–source voltage that is applied to an output transistor. With matched devices, the output current is related to the reference current through the transistor aspect ratios [1], [2].

Generating a current much smaller than the reference ideally requires either a very large device-ratio difference or additional circuitry. The Widlar principle introduces a resistor in the source of the output device, creating a source voltage that reduces the output transistor's effective gate–source voltage and therefore its current. The original Widlar circuit was introduced as a low-value bipolar current source; the same source-degeneration principle applies to MOSFETs [1], [3].

The objective of this work is to design an NMOS Widlar mirror that derives 1 µA from a 100 µA reference, to verify it against the GPDK090 technology models, and — going beyond schematic-level study — to carry the circuit through analog layout, DRC and LVS so that the drawn silicon is proven equivalent to the simulated schematic.

II. CIRCUIT CONFIGURATION AND DESIGN

A. Topology and specifications

The circuit contains two NMOS transistors. M1 is diode-connected with its source grounded, and its gate drives the gate of M2. The source of M2 is returned to ground through the degeneration resistor R_S. A 100 µA DC current source biases the diode-connected branch, while R_{Load} = 1 MΩ connects the output drain node to the 1.8 V supply. Both devices are gpdk090_nmos2v (thick-oxide) transistors with W = 3.12 µm and L = 280 nm. Unlike the ideal resistor commonly used in schematic exercises, R_S here is the PDK polysilicon resistor resnsppoly (400 Ω/sq), so the same element can be drawn in layout. The design values are summarized in Table I.

TABLE I
DESIGN SPECIFICATIONS

Parameter	Value
Supply voltage, VDD	1.8 V
Reference current, IREF	100 µA
Target output current, IOUT	1 µA
Transistor dimensions	W = 3.12 µm, L = 280 nm
Assumed VTH (hand calc)	0.52 V
Assumed µnCox (hand calc)	350 µA/V²
RS realization	resnsppoly poly, 400 Ω/sq
Load resistance, RLoad	1 MΩ

B. Analytical estimate of the source resistance

For an NMOS transistor in saturation the simplified square-law model gives $V_{GS} = V_{TH} + (2I_D/\mu_n C_{ox}(W/L))^{1/2}$. With W/L = 3.12/0.28 = 11.14, the reference branch gives V_{GS1} = 0.52 + 0.226 = 0.746 V and the 1 µA branch gives V_{GS2} = 0.52 + 0.023 = 0.543 V. Applying Ohm's law to R_S: $R_S \approx (V_{GS1} - V_{GS2})/I_{OUT}$ = (0.746 – 0.543)/1 µA ≈ 204 kΩ.

This is an initial estimate only: it neglects body effect on M2 (whose source sits above ground), subthreshold conduction, and short-channel effects, all of which are significant at a 100:1 current ratio.

III. SIMULATION METHODOLOGY

The schematic was implemented in Cadence Virtuoso 6.1.7 (Fig. 1) and simulated with Spectre using the GPDK090 BSIM3v3 models (NN corner). The DC operating point was examined; the source resistance was swept (150–400 kΩ) to select the value that brings the output current to 1 µA; the poly-resistor length was swept to center the drawn geometry; a supply-voltage sweep, an

output-compliance sweep and a 100 µs transient analysis were also recorded. The 1 MΩ load resistor was retained for all simulations shown.

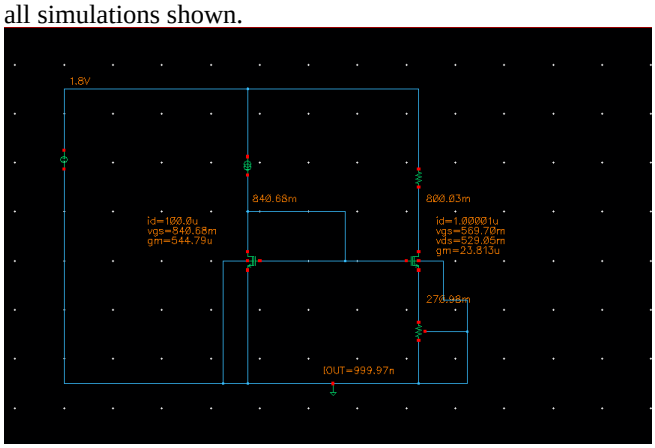


Fig. 1. Cadence Virtuoso schematic with the DC operating point annotated. R_S is the 28-segment resnsppoly resistor (r = 270.98 kΩ).

IV. RESULTS AND DISCUSSION

A. Operating point

At the selected resistance the simulator reports I_D(M2) = 1.00001 µA, V_{GS2} = 569.70 mV, V_{DS2} = 529.05 mV and g_{m2} = 23.81 µS; the source-node voltage is 270.98 mV and the current delivered to the load is 999.97 nA. M1 operates in strong-inversion saturation (V_{GS1} = 840.68 mV). Notably, the simulated V_{TH} of M2 rises to 612 mV through the body effect, so the output device actually operates in moderate/weak inversion — exactly the regime where the square-law hand model fails.

B. Source-resistance sweep

Fig. 2 shows the simulated output current as R_S varies from 150 kΩ to 400 kΩ. The current falls monotonically with increasing resistance, consistent with the source-degeneration mechanism, and crosses the 1 µA target at $R_S \approx 271.0$ kΩ. Table II lists the output current at selected resistances for cross-checking the curve.

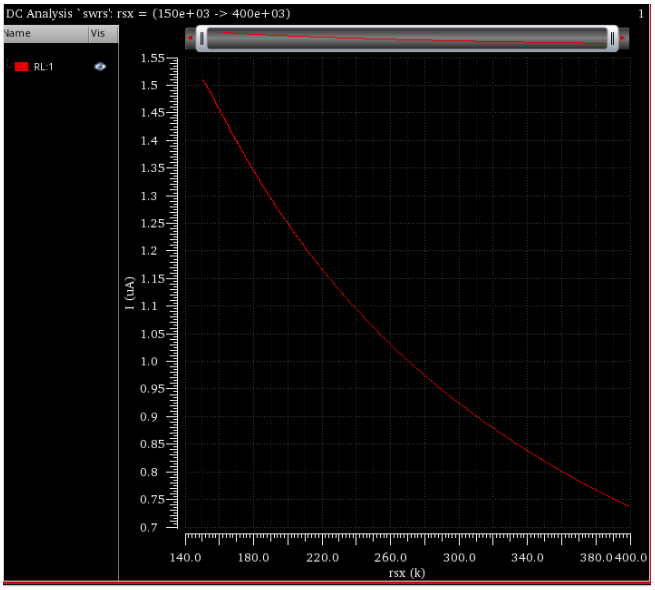


Fig. 2. DC sweep of output current versus source resistance (ViVA). The 1 µA target is reached at $R_S \approx 271.0$ kΩ.

TABLE II
OUTPUT CURRENT AT SELECTED SOURCE RESISTANCES

RS (kΩ)	IOUT (µA)
200	1.249171
225	1.147868
250	1.062244
275	0.988911
300	0.925380
325	0.869789
350	0.820722

The selected resistance is about 33% greater than the 204 kΩ analytical estimate. The discrepancy reflects the difference between the simplified hand model and the technology-model simulation — chiefly the body effect and the moderate-inversion operation of M2. The drawn resistor (28 series segments of 24.195 µm × 1 µm at 400 Ω/sq, total 677.46 µm, 270.98 kΩ) was centered by a separate sweep of the poly length, which places the 1 µA point at 677.44 µm — within 0.003% of the drawn value.

C. Supply-voltage sweep

Fig. 3 shows the DC response as the supply is swept from 0 V to 1.8 V: the output current (upper trace) is established once sufficient headroom exists, while the supply current (lower trace) is dominated by the ≈100 µA reference branch. The output branch requires adequate supply voltage before the intended current is established.

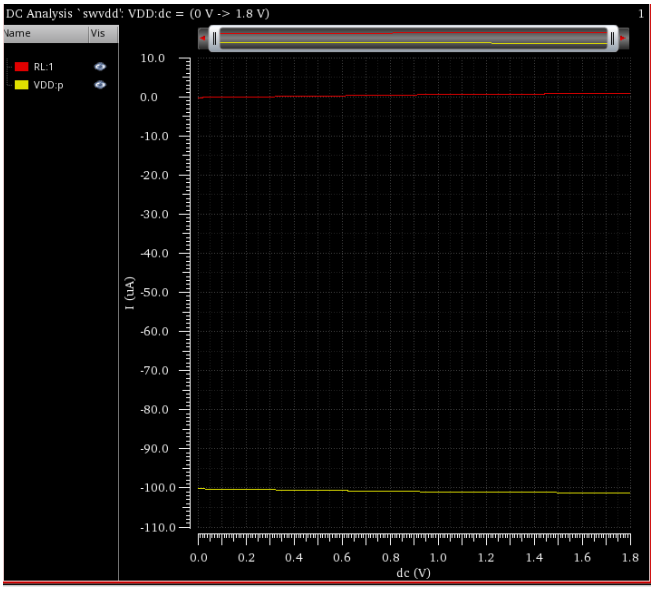


Fig. 3. DC response during the supply-voltage sweep; upper trace is I_{OUT}, lower trace the supply-source current.

D. Output compliance

Replacing the load with a swept voltage source characterizes the mirror as a current source (Fig. 4). The output current saturates at ≈0.9 µA once V_{OUT} exceeds about 0.3 V and rises gently to 1.18 µA at 1.8 V, showing the finite output resistance of the simple Widlar stage. For ±10% current accuracy the output node must be held above roughly 0.3 V.

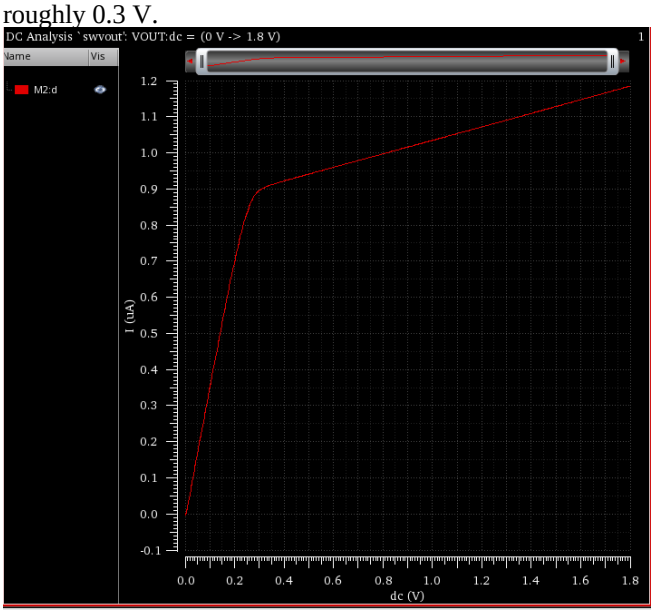


Fig. 4. I_D(M2) versus swept output voltage — output compliance of the mirror.

E. Transient behaviour

A 100 µs transient at the operating point (Fig. 5) shows the output current steady at 999.9745 nA with sub-5 pA numerical ripple, i.e. a stable DC current with no start-up anomaly in this configuration.

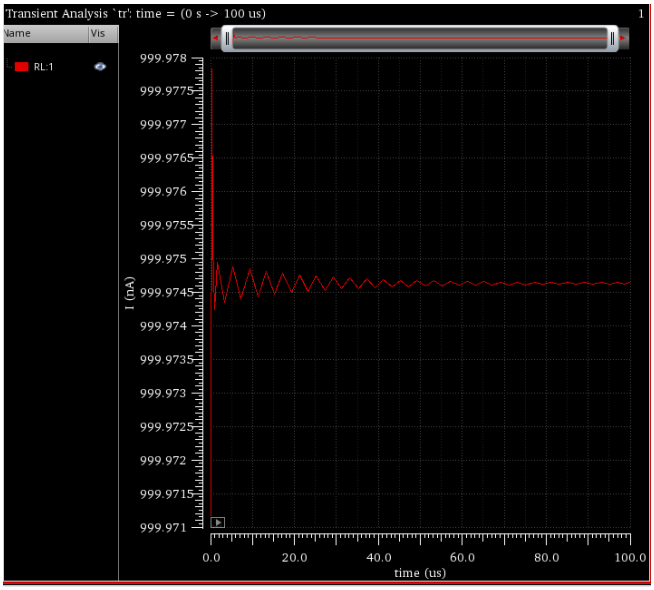


Fig. 5. Transient response of the output current over 100 µs.

V. LAYOUT AND PHYSICAL VERIFICATION

The mirror core (M1, M2, R_S) was laid out in Metal1 in a ≈48 × 37 µm area (Fig. 6), dominated as expected by the 270.98 kΩ serpentine poly resistor. M1 and M2 are single-finger nmos2v pcells with top gate straps (Fig. 7). M1 uses the pcell's integrated body tie — permissible only because its source and bulk are both grounded, as the integrated tie shorts S to B. M2, whose source sits 271 mV above ground, uses no local tap; its bulk is the common substrate, tied to ground by the taps of M1 and of the resistor. The resistor carries its own substrate-tap strip, required by the DRC substrate-stamping rule for its Psub region. The NS net (M2 source to R_S) is routed down the right edge and enters the resistor's PLUS terminal from below the pad row, keeping it clear of every ground shape.

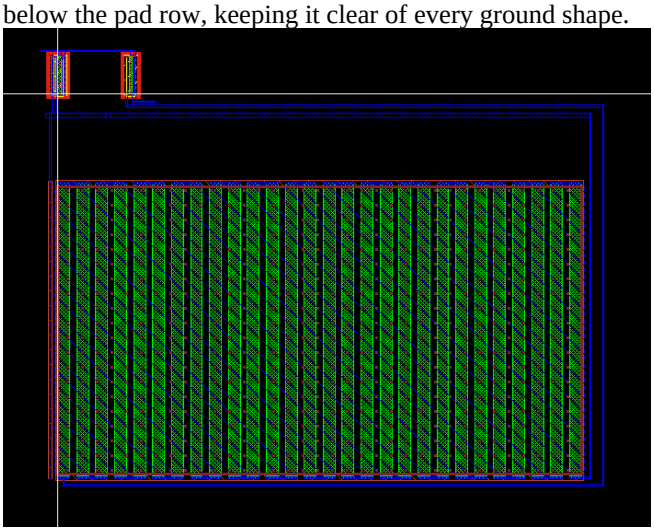


Fig. 6. Complete layout: mirror pair (top left) above the 28-segment serpentine resnsppoly resistor; common ground rail and right-edge NS routing.

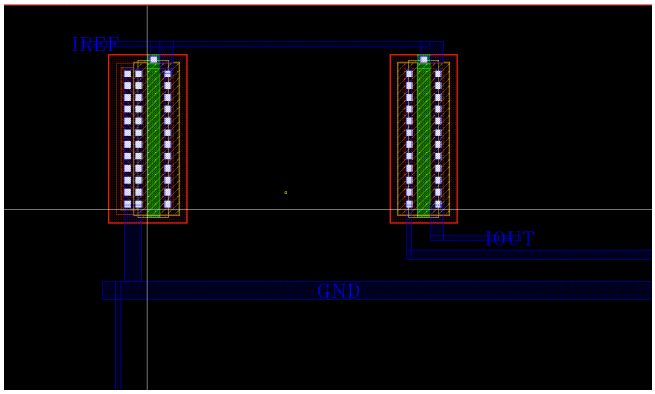


Fig. 7. Detail of the transistor row: M1 (left, with integrated tap) and M2, gate straps, and the IREF / IOUT / GND Metal1 routing.

Physical verification used the Diva decks shipped with GPDK090. DRC reports zero errors. Layout extraction recognizes 2 nmos2v and 28 resnsppoly devices, all well formed. LVS compares the extracted view against the schematic: the rule deck's series-resistor permutation merges the 28 drawn segments into the single 677.46 µm schematic resistor, and the netlists match with zero unmatched instances, nets or terminals and zero size errors (Table III). The layout is therefore electrically identical to the simulated circuit; at DC the residual layout parasitic is ≈87 Ω of Metal1 trace resistance in series with 271 kΩ, a 0.03% effect.

TABLE III

PHYSICAL VERIFICATION SUMMARY

Check	Result
Diva DRC	0 errors
Extraction	2 nmos2v + 28 resnsppoly, 0 errors
LVS devices	30 vs 3 after series merge — matched
LVS verdict	“The net-lists match”, 0 size errors

VI. ADVANTAGES, LIMITATIONS, AND APPLICATIONS

The source resistor provides a simple means of generating a small, stable bias current without extreme device ratios, which is useful in low-current biasing networks [1], [2]. Its accuracy depends on the transistor models, the resistor value and temperature coefficient, the available output voltage, and device mismatch. The resistor dominates the silicon area — here about 95% of the cell — and consumes voltage headroom; the output current departs from target when the output transistor lacks adequate drain voltage, as quantified by the compliance sweep of Fig. 4. Process variation of the poly sheet resistance (±20% across corners) translates directly into output-current variation, a consideration absent from ideal-resistor exercises.

VII. CONCLUSION

An NMOS Widlar current mirror producing 999.97 nA from a 100 µA reference at 1.8 V was designed in Cadence Virtuoso with the GPDK090 library. The square-law hand calculation gave $R_S \approx 204$ kΩ, while the technology-model sweep selected 271.0 kΩ — a 33% correction dominated by body effect and moderate-inversion operation of the output device. The resistor was realized as an on-chip 28-segment poly serpentine, and the complete cell was carried through layout, DRC (clean) and LVS (matched). The exercise demonstrates both the value of analytical estimates refined by simulation with the intended technology models, and the additional constraints — body ties, substrate stamping, routing topology — that only appear at the physical-design stage.

REFERENCES

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[2] Analog Devices, “Chapter 11: The Current Mirror,” Analog Devices Wiki, section on the Widlar current source. [Online]. Available: https://wiki.analog.com/university/courses/electronics/text/chapter-11

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